

Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions

Huy Hoang Le^a, Kim-Anh Nguyen^{b,*}

^a*R&D Department, PowerMore Ltd., Da Nang, 550000, Viet Nam*

^b*Faculty of Electrical Engineering, University of Science and Technology, The University of Danang, Da Nang, 550000, Viet Nam*

Abstract

Health indicators combine condition-monitoring channels through weights fixed at design time, whereas the combination most informative about remaining useful life, the informative direction, rotates when degradation mechanisms progress at different rates. This paper quantifies the information a fixed linear indicator discards and whether the operating point, which governs those rates, can reduce it. For units differing by a time shift, the loss depends only on the angle between the indicator and the informative direction, and inflates the error variance of the remaining-life estimate by an exact factor. If this direction traverses an arc Φ over life, the best fixed indicator inflates the variance by at most $1/\cos^2(\Phi/2)$; on 129 measured battery cells the arc is 30.7° , so by at most 7.5%. On those cells a remaining-life estimator and a replacement policy perform as well with that indicator as with all channels. In simulation, steering the operating point reduces the information loss by a factor of 47, yet a practical estimator using the indicator stays within 1.2% of an estimator using every channel, while lifetime varies 19-fold. Unless the direction rotates widely, a fixed indicator is adequate, and the operating point is better selected for lifetime.

Keywords: health indicator, remaining useful life, prognostics, predictive maintenance, competing degradation mechanisms, operating conditions,

*Corresponding author

Email addresses: `lhhoang@powermore.vn` (Huy Hoang Le), `nkanh@dut.udn.vn` (Kim-Anh Nguyen)

1. Introduction

Condition monitoring seldom supplies raw measurements directly to a prognostic algorithm. Channels such as vibration, temperature, impedance and capacity are first combined into a scalar health indicator, and the prognostic algorithm operates on that scalar alone [1, 2]. The weights of the combination are usually fixed at the design stage, because they are embedded in firmware, frozen by certification, or impractical to re-tune in the field. The indicator therefore determines how accurately the remaining useful life can be estimated and, through that estimate, how conservatively maintenance must be scheduled.

Engineering components commonly degrade through several mechanisms at once, and these mechanisms need not progress at the same rate: one may dominate early in life and saturate, while another accelerates towards failure. Consequently, the combination of channels that carries the most information about the remaining useful life rotates as the component ages, and a fixed indicator can be aligned with it at a single instant only. Information discarded by the indicator cannot be recovered by any subsequent prognostic algorithm [3].

Two questions of practical relevance therefore arise. The first concerns the magnitude of the information lost by a fixed indicator, and whether it can be assessed before the indicator is designed. The second concerns the operating conditions (temperature, load or duty cycle). These vary, either because the duty imposes them or because the operator selects them, and in both cases they alter the relative rates of the mechanisms; the question is whether a selected operating point can keep the indicator aligned and thereby reduce the loss.

Existing work addresses the first question from the perspective of indicator construction, and its criteria differ from the one used here. Health indicators are constructed from data against criteria such as monotonicity and trendability [2, 4], or by fusing sensors into a composite index whose weights are fitted once, for example so that the fused index has a consistent value at failure or follows a prescribed degradation model [5, 6, 7, 8, 9]; such indices have also been built under several operating conditions [10] and learned by neural networks [11], and reviews survey the resulting pipelines [1, 12]. Once fitted, these mappings are fixed and linear in the channels, which is precisely

the class of indicators considered in this paper, but none of the construction criteria measures the information about remaining life that the index discards as the mechanisms rotate. State-space fusion [13] avoids a fixed mapping but tracks a single degradation state, so that no combination of channels can rotate. Section 4 compares one fusion criterion with the design proposed here on measured cells.

The second question connects to degradation modelling and to operation. Degradation under time-varying operating conditions is well modelled [14, 15, 16, 17, 18, 19, 20], and several degradation characteristics are known to respond differently to the same environment [21] or to depend on one another [22]. A separate body of work selects the operating point, or the load or production rate it implies, to extend life or to balance performance against reliability [23, 24, 25, 26, 27, 28], and prognostic accuracy has been assessed through its consequences for predictive maintenance [29]. These studies treat the monitoring signal as given; none asks what the operating point does to the information a fixed indicator retains. Our previous work bounded the information that any fixed compression retains along a given trajectory [3] and the number of run-to-failure units required to learn such a compression [30]; both analyses treat the trajectory as fixed.

The main contributions of this paper are as follows.

- The information discarded by a fixed linear indicator is shown to be an exact function of a single misalignment angle, and the resulting inflation of the error variance of the remaining-life estimate, and of the safety margin of a predictive replacement policy, is obtained in closed form (Section 2).
- The arc traversed by the informative direction over the service life is shown to bound this inflation before any indicator is designed, which yields a five-step design procedure that requires run-to-failure data only (Section 3).
- In a case study on measured battery cells the bound is 7.5%, which corresponds to a safety margin of a predictive replacement policy at most 3.7% wider; a leave-one-out remaining-life estimator and a replacement policy using the fixed indicator perform as well as ones using all channels (Section 4).
- Selecting the operating point to keep the indicator aligned reduces the information loss substantially but leaves the error of a practical

72 remaining-life estimator essentially unchanged, whereas the operating
 73 point alters the lifetime by an order of magnitude (Section 5). Public
 74 run-to-failure data support the rotation premise but do not permit a
 75 test of the effect of the operating point on degradation (Section 6).

76 The remainder of the paper is organised as follows. Section 2 formulates
 77 the degradation model and the information loss. Section 3 derives a bound
 78 on the loss in terms of the rotation arc and the design procedure that fol-
 79 lows from it. Section 4 applies the procedure to measured lithium-ion cells.
 80 Section 5 examines whether selecting the operating point is worthwhile, and
 81 Section 6 examines the premises of the model on public run-to-failure data.
 82 Section 7 discusses the implications and concludes the paper. Derivations
 83 are given in Appendix A.

84 2. Problem formulation

85 2.1. Degradation model with operating-point dependence

86 A component degrades through d mechanisms, with damage states $x =$
 87 $(x_1, \dots, x_d)$, $x_i \geq 0$. A single operating point $u(t) \in \mathcal{U} \subset \mathbb{R}^m$, comprising
 88 variables such as temperature, load and duty cycle within a compact envelope
 89 $\mathcal{U}$, drives all of them:

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d. \quad (1)$$

90 The stress factor $\varphi_i(u) > 0$ expresses the acceleration of mechanism i at u
 91 relative to a reference operating point, and the shape function $h_i(x_i) > 0$
 92 describes the dependence of its rate on the damage already accumulated.
 93 This is the standard accelerated-life form [14]: in the Arrhenius model
 94 $\varphi_i(u) = A_i \exp(-E_i/(kT))$ with activation energy E_i , and suitable choices
 95 of h_i yield exponential, power-law and stretched-exponential degradation [3].
 96 Damage is normalised such that each mechanism exhausts the component at
 97 unity, and the component is withdrawn from service at a condemnation limit
 98 $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (2)$$

99 2.2. Remaining useful life and the informative direction

100 The quantity of interest is the remaining useful life $R(t) = T_f - t$. For a
 101 single unit with known parameters, (1) would determine it exactly; the uncer-
 102 tainty arises from variability between units [14]. The units of the population

are assumed to differ in their position along a common degradation path, that is, by a time shift, and R is taken to be Gaussian over the population with variance $\text{Var}[R]$. Since R decreases at unit rate, the sensitivity of mechanism i to the remaining life is

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (3)$$

Populations whose units differ in their degradation coefficients rather than in their timing are not covered: for the system of Section 5.2, dispersing the coefficients rotates the regression of damage on remaining life by about 45° relative to (3) (Supplementary Section S4). Section 4 examines the practical consequence on measured cells whose lives differ more than fourfold.

Each mechanism is observed through a dedicated channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (4)$$

with independent standard normal noise ϵ_i . This pairing of channels with mechanisms is an assumption: condition monitoring in general yields $y = H(x, u) + \epsilon$, with sensors that mix mechanisms, and Section 6 shows that raw turbofan sensors violate it. Whitening each channel by its noise level gives the signal vector and its direction,

$$\mathbf{d} = [\sigma_i^{-1} \varphi_i(u) h_i(x_i)]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (5)$$

The norm $\|\mathbf{d}\|$ measures the sensitivity of the record to the remaining life, and $\hat{\mathbf{d}}$ is the combination of channels along which this sensitivity acts, referred to as the *informative direction*. All entries of $\mathbf{d}$ are positive, and $\hat{\mathbf{d}}$ depends on both the state and the operating point. The operating point affects $\hat{\mathbf{d}}$ through two paths: indirectly, through the damage it produces, and directly, through $\varphi_i(u)$ (Fig. 1).

2.3. Information loss of a fixed linear indicator

A fixed linear indicator maps the whitened record $\tilde{y} = \Sigma^{-1/2} y$, $\Sigma = \text{diag}(\sigma_i^2)$, through a weight vector selected once at the design stage:

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1. \quad (6)$$

An indicator defined on raw measurements is the same mapping with $\mathbf{v} \propto \Sigma^{1/2} \mathbf{w}$. Let ψ denote the angle between $\mathbf{v}$ and $\hat{\mathbf{d}}$, and $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ the

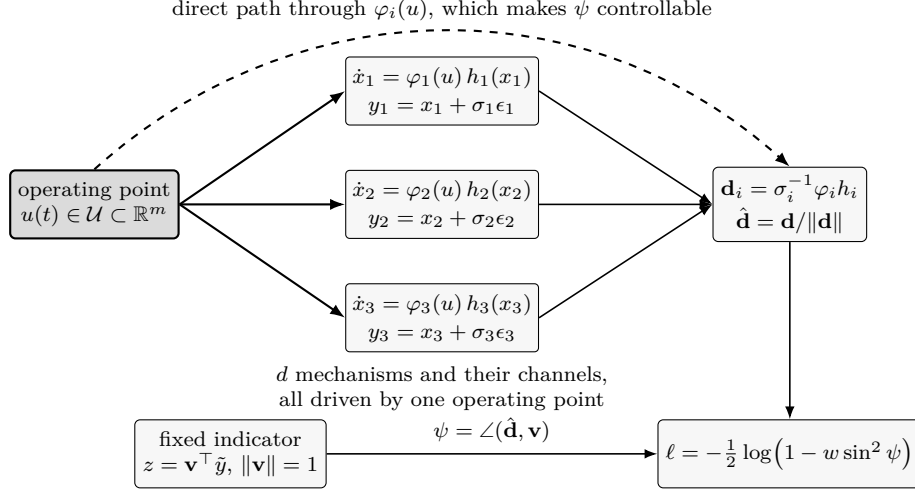


Figure 1: Overview of the problem formulation. The operating point affects the informative direction both through the damage and directly through the stress factors $\varphi_i(u)$; the direct path makes the misalignment ψ controllable. A single operating point drives all d mechanisms.

129 signal-to-noise ratio of the record. Linearising the record about the nominal
 130 path renders it jointly Gaussian with R ; the indicator retains the fraction
 131 $\cos^2 \psi$ of the signal and all of the noise. The information about R that it
 132 discards relative to the complete record is therefore (Appendix A.1)

$$\ell = \frac{1}{2} \log \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (7)$$

133 with $w = \gamma/(1 + \gamma)$, and the error variance of the optimal estimate of R based
 134 on the indicator exceeds that based on the complete record by the factor

$$\frac{\text{Var}[R | z]}{\text{Var}[R | \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}. \quad (8)$$

135 The loss thus depends on the indicator only through a single angle, and for
 136 small misalignment $\ell \approx \frac{w}{2} \psi^2$. The design objective is its average over the
 137 service life,

$$\bar{\ell} = \frac{1}{T_f} \int_0^{T_f} \ell(t) dt. \quad (9)$$

138 The same reduction applies to the information discarded over a complete
 139 record, with γ summed over the measurement instants and $\sin^2 \psi$ averaged
 140 with weights $\|\mathbf{d}\|^2$ (Supplementary Section S2).

141 The factor (8) has a direct consequence for maintenance. Under the
 142 Gaussian posterior of the model, a predictive replacement policy that replaces
 143 a unit once its probability of failing within a lead time h reaches α does so
 144 when the estimated remaining life falls to $h + z_{1-\alpha}\sigma$, where σ is the posterior
 145 standard deviation and $z_{1-\alpha}$ the standard normal quantile. The safety margin
 146 $z_{1-\alpha}\sigma$ is, on average, remaining life discarded at replacement beyond the lead
 147 time, and a fixed indicator multiplies it by e^ℓ relative to the complete record.
 148 Decision-oriented assessments of prognostic algorithms weigh precisely this
 149 margin against the risk of failure [29].

150 Equation (7) also identifies the relevant Fisher information. The design of
 151 accelerated degradation tests and of inputs for estimability usually maximises
 152 the Fisher information of the record [31], here $I_{\text{rec}} = \|\mathbf{d}\|^2$ per unit prior
 153 variance about R ; the loss, however, is determined by this quantity together
 154 with the Fisher information of the indicator itself, $I_z = (\mathbf{v}^\top \mathbf{d})^2$:

$$\ell = \frac{1}{2} \log \left(1 + \text{Var}[R] I_{\text{rec}} \right) - \frac{1}{2} \log \left(1 + \text{Var}[R] I_z \right). \quad (10)$$

155 For a fixed value of I_{rec} the loss still ranges from zero to $\frac{1}{2} \log(1+\gamma)$ as ψ varies.
 156 Maximising the information in the record therefore does not minimise the loss
 157 of a fixed indicator, whereas including I_z in the criterion does. Supplementary
 158 Section S4 illustrates this on the system of Section 5.2.

159 3. Rotation of the informative direction and a bound on the loss

160 3.1. Rotation rate of the informative direction

161 Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i and $p_i = \hat{\mathbf{d}}_i^2$ the
 162 share of the record's Fisher information carried by channel i . Differentiating
 163 (5) along (1) gives (Appendix A.2)

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g}_i = \varphi_i(u) h'_i(x_i) + \nabla_u \log \varphi_i(u) \cdot \dot{u}. \quad (11)$$

164 The informative direction thus rotates at a speed equal to the information-
 165 weighted spread of the logarithmic rates of the mechanisms. If all mecha-
 166 nisms accelerate in proportion, the direction remains fixed however rapidly
 167 the component degrades; rotation arises only from differences between the
 168 mechanisms. The second term of $\dot{g}_i$ represents the influence of the operating
 169 point: a change in the operating point alters the rates that rotate the di-
 170 rection. Only the component of this rotation along the great circle towards
 171 $\mathbf{v}$ changes ψ ; the remaining $d - 2$ components rotate the direction without
 172 affecting the information retained by the indicator (Fig. 2(b)).

173 3.2. A bound in terms of the rotation arc

174 Suppose that the informative direction traverses a total arc $\Phi < \pi$ over
 175 the service life. The direction at the midpoint of this arc lies within $\Phi/2$ of
 176 every point on it, so a fixed indicator placed at the midpoint satisfies $\psi \leq \Phi/2$
 177 throughout. Since the loss increases with ψ , the best fixed indicator satisfies,
 178 at every instant,

$$e^{2\ell} \leq \frac{1}{1 - w \sin^2(\Phi/2)} \leq \frac{1}{\cos^2(\Phi/2)}. \quad (12)$$

179 The bound requires only the arc, which can be estimated from run-to-failure
 180 data before any indicator is designed. An arc of 30° allows at most 7.2% ad-
 181 ditional error variance, 60° at most 33%, and 90° a doubling; the right-hand
 182 bound corresponds to the limit of high signal-to-noise ratio and is indepen-
 183 dent of γ . Correspondingly, the safety margin of the replacement policy of
 184 Section 2.3 grows by at most the factor $1/\cos(\Phi/2)$. The best fixed indicator
 185 lies close to the midpoint of the arc; Supplementary Section S2 characterises
 186 it as a barycentre of the path of the informative direction. For the system
 187 of Section 5.2 under constant inputs, the arc is between 24.8° and 35.6° ,
 188 the bound allows between 4.8% and 10.3%, and the best fixed indicator for
 189 each input incurs between 1.6% and 3.4%; the bound therefore holds and
 190 overestimates the excess by a factor of about three.

191 Figure 2(a) shows the dependence of the loss on the misalignment angle,
 192 and Fig. 2(c) the rotation measured on a battery fleet (Section 4).

193 3.3. Design procedure

194 The results above yield a design procedure that requires run-to-failure
 195 data only.

- 196 1. Estimate the informative direction along normalised life as the time
 197 derivative of the monitoring channels, each whitened by its measure-
 198 ment noise, averaged over the fleet.
- 199 2. Compute the arc Φ that the direction traverses and the bound (12); if
 200 the bound is acceptable, a fixed indicator suffices.
- 201 3. Place the fixed indicator at the barycentre of the path, in practice the
 202 renormalised mean direction.
- 203 4. Validate the indicator with a leave-one-out remaining-life estimator,
 204 against the same estimator run on all channels.
- 205 5. Evaluate the maintenance decision that the estimate supports.

206 Section 4 carries out the five steps on measured lithium-ion cells.

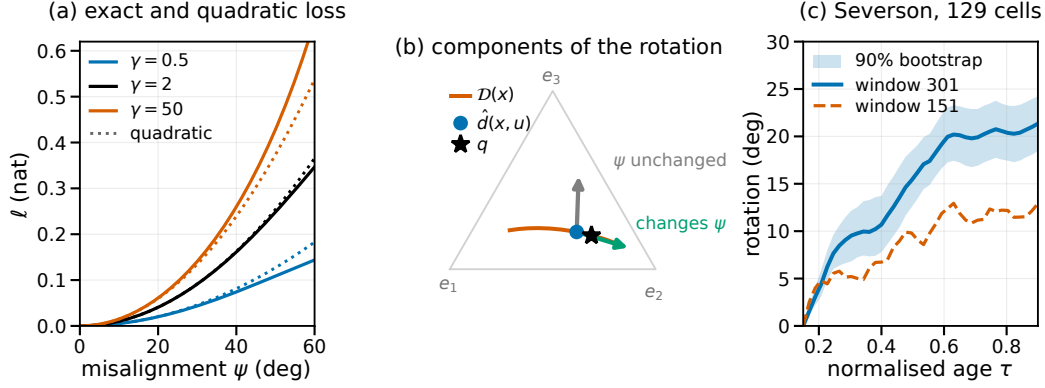


Figure 2: (a) The exact loss (7) (solid) and its quadratic approximation (dotted), whose quartic correction changes sign at $\gamma = 2$. (b) Only the component of the rotation of the informative direction along the great circle towards the indicator changes the misalignment. (c) Rotation measured on the Severson fleet for two differentiation windows.

207 4. Case study: lithium-ion cells

208 The procedure is applied to the lithium-ion cells of Severson et al. [32], cy-
 209 cled to failure under different fast-charging protocols. The monitoring chan-
 210 nels are discharge capacity, internal resistance and mean cell temperature.
 211 The 129 cells that meet the selection criteria of the protocol in Supplementary
 212 Section S1 have lives from 447 to 1932 cycles; because the lives differ more
 213 than fourfold across protocols, the population is far from a time shift, and
 214 the case study also probes that assumption. Supplementary Section S1 gives
 215 the protocol and parameters of every step.

216 4.1. Rotation and the bound

217 The fleet-mean informative direction moves through an arc of 30.7° be-
 218 tween 15% and 90% of life (Fig. 2(c)). By (12), the fixed indicator at its
 219 barycentre therefore inflates the error variance of the remaining-life estimate
 220 by at most 7.5% over that window, irrespective of the signal-to-noise ra-
 221 tio, and widens the safety margin of a predictive replacement policy by at
 222 most 3.7%. In whitened units the indicator weights capacity, resistance and
 223 temperature as (0.99, 0.10, 0.12) and lies at most 14° from the informative
 224 direction.

Table 1: Case study on the Severson cells. $\psi_{\max}$ is the largest angle between the signal and the fleet-mean informative direction over $\tau \in [0.15, 0.90]$; the MSE ratios compare the leave-one-out remaining-life estimator run on the signal with the same estimator run on all three channels, at 50%, 70% and 90% of life; the replacement columns give the minimal long-run cost per 1000 cycles, in units of a preventive replacement, when a failure costs ten times as much, with the fraction of cells failing and the mean life left unused at replacement, in cycles.

signal	$\psi_{\max}$	MSE ratio			replacement policy		
		50%	70%	90%	cost	failures	unused
all channels	—	1	1	1	1.689	1.6%	134
fixed indicator	14°	0.960	0.972	0.999	1.688	1.6%	133
capacity	22°	1.050	1.014	0.986	1.690	1.6%	134
fusion	82°	28.9	8.43	0.807	1.731	1.6%	150
temperature	90°	43.7	171	725	—	—	—

225 4.2. Remaining-life estimation

226 A similarity-based estimator matches the last 30% of a cell’s elapsed cycles
227 against the fleet reference path in normalised life and returns the life that fits
228 best; everything it uses, including the fixed indicator, is computed without
229 the cell being tested. With all three channels its root-mean-square error
230 is 148, 83 and 49 cycles at 50%, 70% and 90% of life. Run on the fixed
231 indicator at the barycentre of the measured direction, the same estimator
232 attains between 0.96 and 1.00 times the mean squared error of the three-
233 channel estimator at these ages (Table 1). Capacity alone, the conventional
234 battery health indicator, lies up to 22° from the informative direction, for
235 which the bound allows 17%, and raises the mean squared error by 5% at half
236 of life and by less later. Temperature alone, a deliberately misaligned control,
237 raises it at least 40-fold, so the test is able to detect a poorly chosen indicator.
238 A data-level fusion indicator, whose weights minimise the between-cell spread
239 of the indicator at failure relative to its mean change over life, in the spirit
240 of [5], lies 75° to 82° from the informative direction and has 28.9 times the
241 mean squared error at half of life. Near failure, the stage its construction
242 targets, it is the most accurate signal, at 0.81 times the three-channel error:
243 it suppresses between-cell differences in the level of the channels at failure, a
244 source of error that the time-shift population of Section 2 does not contain.

245 4.3. Replacement decisions

246 The last step evaluates the decision directly. Each cell is inspected every
247 25 cycles from cycle 100 and is replaced at the first inspection at which the

248 predicted remaining life is at most a threshold θ ; a cell not replaced before its
 249 end of life counts as a failure. For a failure ten times as costly as a preventive
 250 replacement, the threshold that minimises the long-run cost per cycle [29]
 251 gives the costs of Table 1. The fixed indicator, capacity alone and all three
 252 channels reach the same cost to within 0.1%, with 1.6% of cells failing and
 253 about 134 cycles, a fifth of a cell’s life, left unused at replacement; the fusion
 254 indicator costs 2.5% more. For cost ratios from 5 to 50 the fixed indicator
 255 stays within 1% of the three-channel cost. The thresholds are chosen on
 256 the same cells for every signal, so the absolute costs are optimistic but the
 257 comparison is not. The case study thus confirms, on measured data and at
 258 the level of a maintenance decision, what the bound predicts: for this fleet a
 259 fixed indicator is adequate.

260 5. Selecting the operating point

261 The operating point sets the relative rates of the mechanisms and there-
 262 fore moves the informative direction. This section asks whether selecting it
 263 so as to keep a fixed indicator aligned, referred to as steering, is worthwhile.

264 5.1. Limits of steering

265 At a state x , the directions attainable through the operating point form
 266 the set

$$\mathcal{D}(x) = \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\}, \quad (13)$$

267 and the smallest misalignment achievable at x is the floor $\beta_{\min}(x) =$
 268 $\angle(\mathbf{v}, \mathcal{D}(x))$. Steering has two structural limits, derived in Supplementary
 269 Section S4. Holding the misalignment at a constant, non-zero level is a single
 270 scalar condition and needs only one adjustable operating variable. Removing
 271 the loss altogether confines the state to a straight segment in damage space
 272 and, except in special cases, needs one fewer independent operating variable
 273 than there are mechanisms, together with an aligning input that remains
 274 admissible along that segment. A component with three mechanisms and
 275 a single temperature setpoint can therefore reduce the loss but, generically,
 276 not remove it.

277 When the loss cannot be removed, the relevant quantity is the smallest
 278 achievable average loss,

$$\bar{\ell}^* = \min_{u(\cdot)} \frac{1}{T_f} \int_0^{T_f} \ell \, dt. \quad (14)$$

Both the integral and the lifetime T_f depend on the operating point, so the objective is a ratio: the operating point that aligns the indicator is in general not the one that prolongs life. The simplest policy disregards the lifetime and applies, at each state, the input that minimises the misalignment; it is referred to as the myopic policy.

5.2. A numerical study

The study considers $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is a scaled reciprocal temperature, $u = 3(1/(kT) - 1/(kT_{60})) / (1/(kT_{25}) - 1/(kT_{60}))$ with T_{60} and T_{25} the absolute temperatures of 60 and 25 °C, so that $u = 0$ corresponds to 60 °C and $u = 3$ to 25 °C. The parameters are $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, which in these units are activation energies of 0.44, 0.81 and 0.59 eV, with shape functions $h_i(x) = 1 + c_i x$ and $c = (-0.5, 2.0, 0.8)$, which give one decelerating and two accelerating mechanisms. Time is measured in units of the characteristic time of the first mechanism at 60 °C, where $\varphi_1 = 1$. There is a single input $u \in [0, 3]$, so by Section 5.1 the loss can be reduced but, generically, not removed; Supplementary Section S4 confirms this for the present system. The noise has unit variance and the prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of (7) is evaluated at the system's own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$; the values of the reported constants at $x_f = 1$ are given in Supplementary Section S4. The identities used below were verified numerically (Supplementary Section S3).

The myopic policy applies, at each state, the input that minimises the misalignment, searched on a grid of 41 points of $\mathcal{U}$. Minimising its loss over the indicator yields the steering indicator q^* , with weight vector $\mathbf{v}^* = (0.466, 0.771, 0.435)$, at which the myopic policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. This comparison holds the indicator fixed at the one designed for steering. For operation without steering, the indicator would instead be designed for the constant input, and minimising over the indicator and the constant input jointly gives 7.95×10^{-3} , at the high-temperature end $u = 0$ of the envelope. Relative to this baseline, the reduction achieved by steering is a factor of 47. With control grids of 81 points the two factors become 72 and 50, and on 161 points 72 and 50 again. The myopic policy is also close to the best achievable by any policy: the fractional optimum of (14), computed on state grids of up to 137

Table 2: Information loss, remaining-life estimation error and lifetime of the test system under different operating strategies. $\bar{\ell}$ is the information loss (9); $e^{2\bar{\ell}} - 1$ the corresponding excess error variance (8); the RUL errors are the median absolute error of a template-matching remaining-life estimator over 2000 units, as a percentage of life, based on the indicator or on all three channels; T_f is the lifetime. Constant designs use the indicator that is best for their own input.

operating strategy	$\bar{\ell}$	$e^{2\bar{\ell}} - 1$	RUL error (% of life)		
			indicator	all channels	T_f
constant, 60 °C	7.95×10^{-3}	1.6%	10.7%	10.8%	0.378
constant, 41.5 °C	1.44×10^{-2}	2.9%	8.9%	8.8%	1.967
constant, 25 °C	1.26×10^{-2}	2.6%	8.8%	8.7%	7.203
myopic, q^*	1.68×10^{-4}	0.03%	9.5%	9.5%	1.433
myopic, balanced	4.08×10^{-3}	0.8%	8.4%	8.5%	4.928

points per axis, lies within 2.1% of it on every control grid (Supplementary Section S4).

Table 2 reports the practical consequences. By (8), a loss of 7.95×10^{-3} already corresponds to an error variance only 1.6% above that of the complete record, and steering removes almost all of this excess. A practical estimator, however, is largely insensitive to it. A time-shifted population is simulated along the path of each policy and observed at 30 instants over the last 40% of a window with unit channel noise, and the remaining life is estimated from the offset that best matches the reference path of the policy, once from the indicator and once from all three channels. Under every policy the two estimators differ by less than 1.2%, and the error as a fraction of life stays between 8.4% and 10.7%, being determined by the path rather than by the indicator. The operating point does, however, change the lifetime substantially. Across the constant designs of the table it varies 19-fold; the myopic loop at q^* attains a lifetime 1.66 times that of the best constant input at the same indicator and 3.8 times that of the constant design with the smallest loss; and a myopic loop aimed at a balanced indicator, which requires far less steering, extends the lifetime by a further factor of 3.4.

5.3. Sensitivity to model misspecification

An indicator designed from a model inherits the errors of that model. For each level ε , a perturbed energy vector $E(1 + \varepsilon\eta)$ is drawn, with η standard normal, and the barycentre indicator is constructed from this misspecified system along the trajectory of a constant input, as the design procedure of

Table 3: Cost of designing the indicator from a misspecified stress model. Entries are the loss on the true system relative to that of the indicator designed from the true model, over 40 draws per level; ε is the relative standard deviation of the error in the activation energies. The steering policy uses the true system.

ε	median	90th percentile	worst of 40
0.02	1.03	1.52	2.04
0.05	1.40	2.94	6.44
0.10	3.10	10.8	67.7
0.20	10.0	44.9	78.6
0.35	36.0	94.1	195

338 this paper prescribes. The indicator is then evaluated on the true system
 339 under the myopic policy, with every loss evaluated at the system’s own γ .
 340 Since the policy is computed from the true system, the table isolates the cost
 341 of an error in the indicator; an error that also affected the controller would
 342 be more costly. Table 3 reports the ratio of this loss to that of the indicator
 343 designed from the true model, over 40 draws per level; a ratio of 1 indicates
 344 that the error has no cost.

345 An error of 2% in the energies has a negligible cost. At 5% the median
 346 cost is a factor 1.40. At 10% the median cost is 3.10 and the worst of forty
 347 draws costs 67.7, comparable in magnitude to the entire benefit of steering;
 348 beyond this level every column increases monotonically, to a median of 36.0
 349 at $\varepsilon = 0.35$. The loss to which these ratios apply is small, however, so that
 350 even a tenfold increase leaves the indicator’s excess error variance below 1%.

351 6. Evidence from public run-to-failure data

352 Three premises of the model can be tested on public run-to-failure data:
 353 that the informative direction rotates; that the operating point moves it by
 354 acting on the degradation; and that the separable stress model, interpreted
 355 through the pairing of channels with mechanisms, holds. Table 4 summarises
 356 what six public run-to-failure fleets establish and what they leave open. Sup-
 357plementary Section S1 gives the protocol behind every row, and its Table S1
 358 lists the parameters of the code that produced it.

359 Rotation of the informative direction is observed on the battery cells
 360 of Section 4. That analysis and the turbofan contrasts below interpret a
 361 time derivative as the sensitivity (3), and therefore presuppose the time-shift
 362 population of Section 2.

Table 4: Premises of the model examined on six public run-to-failure fleets. Only the two turbofan simulations vary the operating point within a unit, and neither allows it to act on the damage; the actuation premise is therefore untested on every public fleet.

fleet	premise	quantity measured	finding
Severson lithium-ion [32]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); remains in the open positive orthant once the window reaches 301 cycles
XJTU-SY [33], PRONOS-TIA [34]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [35]	actuation	—	only four cells
C-MAPSS [36]	actuation: not testable	between-regime minus within-regime angle, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; surrogate regimes give $ B - W \leq 0.5^\circ$; the damage is independent of the regime by construction, so the contrast reflects the sensor map
N-CMAPSS DS02, DS01 [37]	separability with pairing (4)	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected jointly

363 The effect of the operating point on degradation remains untested. The
 364 measured fleets either do not vary the operating point within a unit or con-
 365 tain too few units. The two turbofan simulations do vary it, but their gen-
 366 erators preclude the test. C-MAPSS imposes its flow and efficiency losses
 367 as exponential functions of time, from a random initial deterioration, and
 368 lets the regime act only on the thermodynamic response of the sensors [36].
 369 N-CMAPSS imposes a linear normal degradation followed by a steeper ab-
 370 normal one, with the operating history entering through the onset of the
 371 abnormal phase [38], and the health parameters supplied with the data are
 372 constant within every flight. The C-MAPSS contrast, positive in all but three
 373 engines against a negative control that returns none, therefore shows that the

374 observed direction depends on the regime through the sensor response alone.
 375 Variable operating conditions thus rotate the observed direction even when
 376 they do not act on the damage. Because the loss of a fixed indicator depends
 377 on the observed direction regardless of the cause of its motion, this finding is
 378 relevant to indicator design; it is not, however, evidence that the operating
 379 point acts on the damage.

380 The separable model fails on raw turbofan sensors. On both N-CMAPSS
 381 datasets the interaction between operating point and life stage exceeds the
 382 main effect of the operating point, and since the simulated damage does not
 383 respond to the operating point within a flight, the interaction is attributable
 384 to the sensor map: the sensors mix mechanisms with weights that depend
 385 on the operating point, in violation of (4). The pairing is what makes the
 386 informative direction positive, so the geometry of Sections 3 and 5 applies
 387 to such sensors only after the observation map has been inverted, and the
 388 offline computation of the actuation authority described in Supplementary
 389 Section S4 is available only where the separable form has been shown to hold.
 390 The two-way layout, whose protocol is given in Supplementary Section S1,
 391 provides this test at the cost of a single pass over data that a reliability
 392 programme typically already holds.

393 7. Discussion and conclusions

394 The results have two practical implications.

395 First, the rotation arc should be assessed before an adaptive indicator is
 396 considered. The loss of a fixed indicator is governed by the extent to which
 397 the informative direction rotates, and (12) converts the arc into a worst-case
 398 increase in the error variance of the remaining-life estimate. For the battery
 399 fleet examined here the resulting increase is at most 7.5%, and a remaining-
 400 life estimator and a replacement policy using the fixed indicator perform as
 401 well there as ones using all channels; for the synthetic system the actual
 402 excess is about one third of the bound. Where the arc amounts to a few
 403 tens of degrees, a single fixed indicator is adequate; adaptive, segmented or
 404 learned indicators become worthwhile as the arc approaches a right angle,
 405 at which the bound allows the error variance to double. The five steps of
 406 Section 3.3 make this assessment from run-to-failure data alone.

407 Second, the operating point should be selected primarily for service life.
 408 Steering the operating point can reduce the information loss by an order of
 409 magnitude or more. Where the loss is already small, however, this yields

almost no improvement in the accuracy of a practical remaining-life estimator, whereas the same choice alters the lifetime by an order of magnitude. Aligning an indicator and prolonging the life of the component are distinct objectives, and the latter is usually the more consequential.

Limitations. Four limitations qualify these conclusions. First, the construction assumes a time-shift population, whereas dispersing the degradation coefficients of the test system rotates the population regression of damage on remaining life by about 45° from the direction the construction uses, more than the measured arc of the battery cells, so that the bound does not cover such dispersion even though the battery case study suggests that the practical conclusion survives it. Second, the pairing of channels with mechanisms fails on raw turbofan sensors, so the geometry applies to them only after the observation map has been inverted. Third, the actuation premise remains untested on every public fleet. Finally, the best achievable loss is computed and bracketed between the myopic policy and an extrapolation, not bounded analytically.

Appendix A. Derivations

Appendix A.1. Information loss and error-variance inflation

Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$, the linearised whitened record. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$. The record's sufficient statistic $\mathbf{d}^\top \tilde{y}$ has ratio γ . A scalar Gaussian channel of signal-to-noise ratio s carries $\frac{1}{2} \log(1 + s)$ nats about R [39], and subtracting gives the first form of (7); substituting $\cos^2 \psi = 1 - \sin^2 \psi$ gives the second. The posterior variances are $\text{Var}[R]/(1 + \gamma \cos^2 \psi)$ and $\text{Var}[R]/(1 + \gamma)$, whose quotient is (8). Equation (10) is (7) with $\gamma = \text{Var}[R]I_{\text{rec}}$ and $\gamma \cos^2 \psi = \text{Var}[R]I_z$; at fixed I_{rec} , w is fixed and ℓ increases continuously from zero towards $\frac{1}{2} \log(1 + \gamma)$ as ψ goes from 0 to $\pi/2$.

Appendix A.2. Rotation of the informative direction

From $\mathbf{d}_i = e^{g_i}$, $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, and $\dot{\hat{\mathbf{d}}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components $\hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$ with $p_i = \hat{\mathbf{d}}_i^2$, so its squared norm is $\text{Var}_p[\dot{g}]$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (1) gives $\dot{g}_i$ in (11). With $\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v}$, $-\sin \psi \dot{\psi} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$, so only the component of $\dot{\hat{\mathbf{d}}}$ along the great circle towards $\mathbf{v}$ changes ψ .

444 **Data availability and reproducibility**

445 The six fleets named in Section 6 are publicly available. One script re-
446 computes every reported number from a single configuration and compares it
447 with the value stated in this manuscript and in its supplementary material,
448 covering the tables and the quantities reported alongside them; all of its
449 checks pass, and the output of the run that verified this version is included.
450 The random-system sweep and the downstream repetition of Supplemen-
451 tary Section S4 and the grid sequence of Section 5.2 require several hours
452 of computation, so they are recorded by three further scripts; the checks
453 read these records and recompute part of the grid sequence afresh. One
454 group of numbers lies outside the checks: the residuals of the identity checks
455 in Supplementary Section S3. The figures are produced by three further
456 scripts. All of these are available at [https://github.com/HuyHoangLe0201/](https://github.com/HuyHoangLe0201/health-indicator-information-loss)
457 [health-indicator-information-loss](https://github.com/HuyHoangLe0201/health-indicator-information-loss), and are also provided with this ar-
458 ticle as a supplementary archive, together with the Supplementary Material.

459 **CRedit authorship contribution statement**

460 **Huy Hoang Le:** Conceptualization, Methodology, Formal analysis, Soft-
461 ware, Validation, Investigation, Data curation, Visualization, Writing – orig-
462 inal draft. **Kim-Anh Nguyen:** Conceptualization, Methodology, Valida-
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464 **Declaration of generative AI and AI-assisted technologies in the** 465 **manuscript preparation process**

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